

The Hamiltonian $H = (-1/2)d^2/dx^2 + x^2/2 + \lambda/x^2$ reobserved*

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The Schrödinger problem for the title Hamiltonian is considered as a perturbed one-dimensional harmonic oscillator. Exact bound state solutions can be derived from a classical differential equation in the theory of Laguerre polynomials. These solutions are valid and analytically dependent on λ only in a limited range of the perturbation strength. Within this region the oscillator Hamiltonian restricted to odd and even parity subspaces is unitary equivalent to H restricted over the respective perturbed subspaces. It is shown that due to the singular nature of the perturbation the allowed λ range is narrowed if side conditions are imposed to make the wavefunctions "physically interpretable."

1. EXACT SOLUTION OF THE PERTURBED PROBLEM IN $L^2(-\infty, +\infty)$

The differential operator H of the eigenvalue problem

$$H\psi(x) = E\psi(x), \quad (1)$$

where $\psi(x)$ belongs to $L^2(-\infty, +\infty)$, can conveniently be written in the form

$$H = H_0 + \lambda V, \quad (2)$$

$$H_0 = -\frac{1}{2}d^2/dx^2 + x^2/2, \quad V = 1/x^2.$$

The unperturbed Hamiltonian H_0 is the harmonic oscillator operator while V is a singular perturbation the strength of which is measured by the parameter λ . For convenience I have set $\hbar = 1$ and considered a particle of unit mass. The unperturbed eigenvalue problem, obtained from (1) in the limit $\lambda \rightarrow 0$, has the well-known solutions

$$E_n^0 = n + \frac{1}{2}, \quad \psi_n^0(x) = (2^n n! \sqrt{\pi})^{-1/2} e^{-x^2/2} H_n(x), \quad (3)$$

where n is a nonnegative integer and $H_n(x)$ are the Hermite polynomials. The solutions of (1) can be obtained directly if one identifies the problem with the following differential equation occurring in the theory of orthogonal polynomials (Ref. 1):

$$-\frac{1}{2} \frac{d^2 \varphi_n^{(\alpha)}(x)}{dx^2} - \left(2n + \alpha + 1 - \frac{x^2}{2} + \frac{1 - 4\alpha^2}{8x^2} \right) \varphi_n^{(\alpha)}(x) = 0, \quad (4)$$

which, when $\alpha > -1$, has solutions in $L^2(0, +\infty)$ given by

$$\varphi_n^{(\alpha)}(x) = x^{\alpha+1/2} e^{-x^2/2} L_n^{(\alpha)}(x^2), \quad (5)$$

where the $L_n^{(\alpha)}$ are the standard generalized Laguerre polynomials. Identifying the differential operator in (4) with H , one finds an expression for the perturbation strength

$$\lambda = \frac{1}{2}\alpha^2 - \frac{1}{8}. \quad (6)$$

Since λ depends only on α^2 , the eigenvalues and eigenfunctions will occur in pairs associated with $+\alpha$ or $-\alpha$, where α is given by

$$\alpha = +\frac{1}{2}\sqrt{1 + 8\lambda}. \quad (7)$$

From (4) one sees that the eigenvalues are

$$E_n^{(\pm\alpha)} = 2n \pm \alpha + 1 \quad (8a)$$

with corresponding eigenfunctions, which are either odd or even since H is parity invariant, given by

$$\psi_n^{(\pm\alpha)}(x) = [n!/\Gamma(n \pm \alpha + 1)]^{1/2} \begin{cases} \left(\begin{smallmatrix} + \\ + \end{smallmatrix} \right) \varphi_n^{(\pm\alpha)}(|x|), & x \geq 0, \\ \left(\begin{smallmatrix} - \\ + \end{smallmatrix} \right) \varphi_n^{(\pm\alpha)}(|x|), & x < 0. \end{cases} \quad (8b)$$

Such a pair of solutions exists only if α belongs to the half open interval $[0, +1[$ since for $\alpha = +1$ the eigenfunction $\psi_n^{(-\alpha)}(x)$ is no longer square integrable. The following orthogonality relations hold:

$$\int_{-\infty}^{+\infty} \bar{\psi}_n^{(\epsilon\alpha)}(x) \psi_{n'}^{(\epsilon'\alpha)}(x) dx = \delta_{nn'} \delta_{\epsilon\epsilon'}, \quad (9)$$

where ϵ and ϵ' may equal ± 1 . For $\epsilon = \epsilon'$ this follows from the orthogonality relations of Laguerre polynomials while for $\epsilon \neq \epsilon'$ (9) holds for parity reasons, as can be verified directly from (5).

The function $\alpha = \alpha(\lambda)$ has a branch point at $\lambda = \frac{1}{8}$. Introducing a cut from $-\frac{1}{8}$ to $-\infty$ in the complex λ plane, an analytic continuation of α can be realized on a double Riemann sheet. Taking the limit $\lambda \rightarrow 0$ along a certain path on this surface, we have that

$$\lim_{\lambda \rightarrow 0} E_n^{(-\alpha)} = 2n + \frac{1}{2} = E_{2n}^0, \quad (10)$$

$$\lim_{\lambda \rightarrow 0} E_n^{(+\alpha)} = 2n + \frac{3}{2} = E_{2n+1}^0,$$

i.e., the eigenvalues $E_n^{(-\alpha)}$ and $E_n^{(+\alpha)}$ are the perturbed energies corresponding to the unperturbed even and odd labelled levels respectively. A similar result holds for the eigenfunctions which are analytical functions of λ on the Riemann sheet as can be seen from (5). It follows that

$$\lim_{\lambda \rightarrow 0} \psi_n^{(-\alpha)}(x) = (-)^n \psi_{2n}^0(x), \quad (11)$$

$$\lim_{\lambda \rightarrow 0} \psi_n^{(+\alpha)}(x) = (-)^n \psi_{2n+1}^0(x),$$

which is easily checked using the relations between Hermite and Laguerre polynomials:

$$\begin{aligned} H_{2n}(x) &= (-)^n 2^{2n} n! L_n^{(-1/2)}(x^2), \\ H_{2n+1}(x) &= (-)^n 2^{2n+1} n! x L_n^{(+1/2)}(x^2). \end{aligned} \quad (12)$$

According to (8a) the perturbed energy spectrum of (1) consists of two sets of equidistant levels $E_n^{(\pm\alpha)}$ which can be obtained from the unperturbed spectrum by uniformly shifting the even (odd) parity levels over a distance $\Delta = -\alpha + \frac{1}{2}(-\Delta)$. The corresponding nonidentical twin eigenfunctions $\psi_n^{(-\alpha)}(x)$ and $\psi_n^{(+\alpha)}(x)$ exist only for α in the half open interval $[0, +1[$. From (6) one can then immediately derive the allowed range for the perturbation strength. It follows that λ has to be situated in the half open interval $[-\frac{1}{8}, +\frac{3}{8}[$. These bounds have a very precise mathematical meaning, in fact, the interval $[-\frac{1}{8}, +\frac{3}{8}[$ is nothing but the neighborhood of $\lambda=0$ within which the solutions depend analytically upon the perturbation strength. The upper limit $\lambda = +\frac{3}{8}$, obtained for $\alpha = +1$, gives the maximum value of λ for which the solutions of (1) can be represented by (5). Indeed the wavefunction $\psi_n^{(-\alpha)}(x)$ ceases to be square integrable at $\alpha = +1$. Therefore, the form of these wavefunctions must change in a singular way at $\lambda = +\frac{3}{8}$. This is also reflected in the fact that the limit of the ground state is zero ($\lim_{\lambda \rightarrow 0} E_0^{(-\alpha)} = 0$) which is clearly inconsistent since the Hamiltonian is positive definite for $\lambda = +\frac{3}{8}$. At the lower limit for the perturbation strength $\lambda = -\frac{1}{8}$ the perturbation is negative definite. It can be shown from a theorem by Kato (Ref. 2) (as has been mentioned also in Ref. 3) that this number is the minimum value for which H is still bounded from below. Physically this is associated with the possibility for the particle to "fall" to the center, i.e., the occurrence of infinite negative energies (Ref. 4). Here the point $\lambda = -\frac{1}{8}$ showed up in a mathematical way as a branch point beyond which the solution of (1) (eigenfunctions and eigenvalues) is no longer analytically dependent on λ . These bounds for the perturbation strength can also be obtained from the solutions of (1) near the origin. One finds that the characteristic exponents (which describe the singularities of the solutions near $x=0$) are given by $(1 \pm \sqrt{1+8\lambda})/2$. It is a well-known fact (Ref. 5) that second order differential operators are no longer bounded from below if the characteristic exponents are imaginary, i.e., if $\lambda < -\frac{1}{8}$. Furthermore, if $\lambda \geq +\frac{3}{8}$, the solution corresponding to the minus sign is not square integrable at the origin. In fact, $\lambda = +\frac{3}{8}$ is the transition point from limit circle to limit point case in the Weyl classification of second order differential equations (Ref. 6). From a theorem by Titchmarsh (Ref. 7) one can show that for $\lambda \in [-\frac{1}{8}, +\frac{3}{8}[$ the differential operator is of the limit circle type at $x=0$ (i.e., there exist two linearly independent solutions square integrable at $x=0$) while for $\lambda \geq +\frac{3}{8}$ one is in the limit point case (i.e., there is a unique solution square integrable at $x=0$).

In Refs. 3 and 8 the solutions $\psi_n^{(-\alpha)}(x)$ were considered to be improper eigenfunctions, in spite of their square integrability, due to their behavior at the origin (see, further, Sec. III). In order to get a complete set of eigenfunctions, one considered as "physically acceptable solutions" on the real line the functions $\varphi_n^{(+\alpha)}(x)$ with the prescription

$$\varphi_n^{(+\alpha)}(-x) = \pm \varphi_n^{(+\alpha)}(x), \quad x \geq 0. \quad (13)$$

The resulting eigenfunctions are $\psi_n^{(+\alpha)}(x)$ and a new function $\phi_n^{(+\alpha)}(x)$, which is energywise degenerate with $\psi_n^{(+\alpha)}(x)$. However, this degeneracy is not removed in the limit $\lambda \rightarrow 0$ while the unperturbed even parity levels do not show up at all. Therefore, one is in fact treating another eigenvalue problem, corresponding to an operator defined by the integral kernel

$$\begin{aligned} H(x, x') &= \sum_{n=0}^{\infty} E_n^{(+\alpha)} (\psi_n^{(+\alpha)}(x) \bar{\psi}_n^{(+\alpha)}(x') \\ &+ \phi_n^{(+\alpha)}(x) \bar{\phi}_n^{(+\alpha)}(x')) \end{aligned} \quad (14)$$

rather than the eigenvalue problem (1). Observe that the limit of (14) as $\lambda \rightarrow 0$ is only "half an harmonic oscillator" (odd parity levels only) with twofold degenerate levels (Ref. 3).

II. UNITARY EQUIVALENCE

The question of unitary equivalence which was raised in Ref. 3 can now be treated as follows. Let $H_0^{(+)}$ be the restriction of H_0 to the even parity subspace $(*)L^2(-\infty, +\infty)$ spanned by the unperturbed eigenstates $\{\psi_{2n}^0(x)\}$. Similarly we introduce $H^{(-\alpha)}$ as the restriction of H over the subspace generated by $\{\psi_n^{(-\alpha)}(x)\}$. The unitary mapping $U(-\alpha)$ which maps the above subspaces onto each other immediately establishes a unitary equivalence between $H^{(-\alpha)} - \Delta I^{(-\alpha)}$ and $H_0^{(+)}$ ($I^{(-\alpha)}$ is the unit operator in the space spanned by $\{\psi_n^{(-\alpha)}(x)\}$). Indeed it is easily checked that

$$[U(-\alpha)H_0^{(+)}U^*(-\alpha) - (H^{(-\alpha)} - \Delta I^{(-\alpha)})]\psi_n^{(-\alpha)}(x) = 0, \quad (15)$$

where Δ is the level shift. In the limit $\lambda \rightarrow 0$, $U(-\alpha)$ equal $I^{(-1/2)}$ and (15) becomes a trivial identity. In the same way one can prove that $H_0^{(-)}$ is unitary equivalent to $H^{(+\alpha)} - \Delta I^{(+\alpha)}$ (with obvious notations for the operator). The existence of the properties of the restricted Hamiltonians demonstrated above is not very surprising. Indeed, the perturbation causes a uniform shift Δ ($-\Delta$) of even (odd) parity levels. The unitary equivalence of $H^{(-\alpha)}$ and $H_0^{(+)}$ ($H^{(+\alpha)}$ and $H_0^{(-)}$) is nothing but the operator analog of this special relation between the unperturbed and perturbed spectrum.

III. CONSEQUENCES OF PROBABILISTIC INTERPRETATIONS OF NONRELATIVISTIC QUANTUM MECHANICS

The original formulations of nonrelativistic quantum mechanics (QM) given by Heisenberg, Born, Jordan (matrix mechanics) and Schrödinger (wave mechanics) were shown to be mathematically equivalent by the latter one (Ref. 9). This was accomplished essentially by proving that the sequential Hilbert space is isometric to the space of square integrable functions. A little later Dirac (Ref. 10) managed to formulate QM in terms of elements of an abstract Hilbert space. In fact, he showed that any realization of the abstract properties of Hilbert space in different mathematical forms gives a possible formulation of QM. Therefore, as is, for instance stated by von Neumann (Ref. 11), the only restriction on the wavefunction in the Schrödinger picture is its single-valuedness and its

square integrability. The integral $\int_a^b |\psi(x)|^2 dx$ is then to be interpreted as the probability of finding the particle in the interval $[a, b]$. However, some authors (see, e.g., Refs. 8 and 12) consider a wavefunction physically interpretable only if $\psi(x)$ and $\psi(x)d\psi(x)/dx$ are continuous. These conditions assure that the point probability density $|\psi(x)|^2$ and the probability density current $\text{Re}[\bar{\psi}(x)(\nabla/i)\psi(x)]$ vary continuously. In the case of singular potentials these conditions turn out to be too severe since one might lose physically observed states (e.g., the ground state of the hydrogen atom).

An interesting illustration of the implications of continuity conditions can now be given for the eigenvalue problem (1). The solutions as derived in Sec. I correspond to the most general point of view, namely the solution in $L^2(-\infty, +\infty)$ (notice that all spectra are entirely discrete such that no difficulties occur due to continuous parts in the spectra). If one restricts oneself to continuous wavefunctions, it follows from (5) that α has to be smaller than $+\frac{1}{2}$, which gives an allowed range of $[-\frac{1}{8}, 0[$ for the perturbation strength. This means that continuous wavefunctions exist only for not too strong negative definite perturbations. If in addition one requires that $\psi(x)d\psi(x)/dx$ is continuous, one finds that $\alpha=0$ such that $\lambda=-\frac{1}{8}$ would give the only possible perturbed Hamiltonian which would be isolated from the unperturbed $\lambda=0$ problem. As an alternative one may choose not to consider the $\psi_n^{(-\alpha)}(x)$ solutions at all (Refs. 3, 8) but then, as was shown in Sec. I, one is no longer considering the same eigenvalue problem.

The final justification for the conditions one imposes on the wavefunctions is, of course, the agreement of

theory and experiment. The analysis given above merely shows that in the case of singular interactions too severe restrictions may have drastic consequences.

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